Reinforcement Learning for Optimal Setpoint Determination in Energy Utility Systems

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Energy utility systems is the backbone of modern energy infrastructures. Their complexity ranges from small, localized systems to vast, highly interconnected infrastructure networks. Most of them operate under dynamic conditions where multiple interacting variables influence operational efficiency, cost, reliability, and stability. Conventionally, optimal setpoints for such systems are determined using mathematical optimisation. These approaches rely on accurate system models and are typically formulated as constrained optimisation problems. While effective for well-defined and moderately sized systems, classical optimisation becomes computationally expensive as system complexity increases. Furthermore, time-varying dynamics, uncertainty from renewable source integration, and model-plant mismatches challenge the applicability of purely model-based approaches.

The central problem addressed in this thesis is whether reinforcement learning can reliably and efficiently determine optimal setpoints for two distinct systems: a simple battery energy storage system and a more complex and non-linear cookie production factory. The challenge arises from the need to simultaneously minimise operational costs, maintain constraint satisfaction, and ensure stable system behaviour under uncertainty. Classical optimisation methods struggle with scalability and adaptability, particularly in dynamic environments where conditions change continuously. Reinforcement Learning (RL) has emerged as a promising data-driven paradigm that learns optimal decision policies through interaction with the environment, without requiring explicit analytical system models. Despite its success in robotics and continuous benchmarks, the application of RL to complex energy utility systems remains insufficiently explored. This thesis proposes to formulate the setpoint optimisation problem as a Markov Decision Process (MDP) and to develop a reinforcement learning framework capable of learning adaptive policies. By comparing RL-based approaches with established optimisation methods, the research aims to identify strengths, weaknesses, and application boundaries of both approaches.

The evaluation of the proposed method will be conducted through systematic simulation studies on two energy system case studies of varying complexity. Performance will be assessed using quantitative metrics including total operational cost, constraint violation frequency, and computational time. The results will provide a structured comparison between reinforcement learning and classical optimisation and will determine under which conditions RL offers practical advantages for real-time setpoint optimisation in complex energy utility systems.